

Current Electricity

Chapter [3]

Class - XII

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Limitations of Ohm's Law

- Ohm's law has been found valid over a large class of materials.

Limitations of Ohm's Law

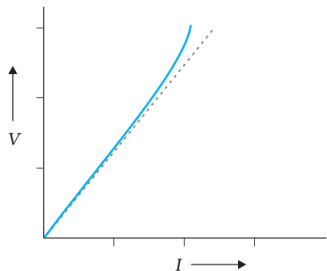


FIGURE 3.5 The dashed line represents the linear Ohm's law. The solid line is the voltage V versus current I for a good conductor.

- But, there do exist materials and devices in which Ohm's law doesn't hold (*i.e.*, $V \propto I$)
 - a V is not proportional to I for some conductor at high voltage.

Limitations of Ohm's Law

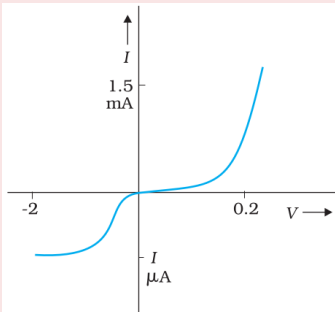
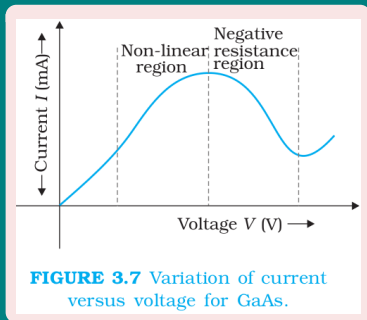


FIGURE 3.6 Characteristic curve of a diode. Note the different scales for negative and positive values of the voltage and current.

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 - a V is not proportional to I for some conductor at high voltage.
 - b For semiconductors (diodes) the relation between V and I depends on the sign of V .

Limitations of Ohm's Law



- But, there do exist materials and devices in which Ohm's law doesn't hold (*i.e.*, $V \propto I$)
 - a V is not proportional to I for some conductor at high voltage.
 - b For semiconductors (diodes) the relation between V and I depends on the sign of V .
 - c The relation between V and I is not unique, *i.e.*, there is more than one value of V for the same current I . A material exhibiting such behaviour is GaAs.

RESISTIVITY OF VARIOUS MATERIALS

Material	Resistivity, ρ ($\Omega \text{ m}$) at 0°C
Conductors	
Silver	1.6×10^{-8}
Copper	1.7×10^{-8}
Aluminium	2.7×10^{-8}
Tungsten	5.6×10^{-8}
Iron	10×10^{-8}
Platinum	11×10^{-8}
Mercury	98×10^{-8}
Nichrome (alloy of Ni, Fe, Cr)	$\sim 100 \times 10^{-8}$
Manganin (alloy)	48×10^{-8}
Semiconductors	
Carbon (graphite)	3.5×10^{-5}
Germanium	0.46
Silicon	2300
Insulators	
Pure Water	2.5×10^5
Glass	$10^{10} - 10^{14}$
Hard Rubber	$10^{13} - 10^{16}$
NaCl	$\sim 10^{14}$
Fused Quartz	$\sim 10^{16}$

The materials are classified as **conductors**, **semiconductors** and **insulators** depending on their resistivities.

- **Conductors (Metals)** have **low resistivities** in the range of $10^{-8} \Omega\text{m}$ to $10^{-6} \Omega\text{m}$.
- **Insulators** like ceramic, rubber and plastics have **high resistivities**. $\approx 10^{18}$ times greater than metals or more.
- **semiconductors** (like Silicon, Germanium etc) have resistivities **in between** the **Conductors** and **Insulators**.

Types of Resistors

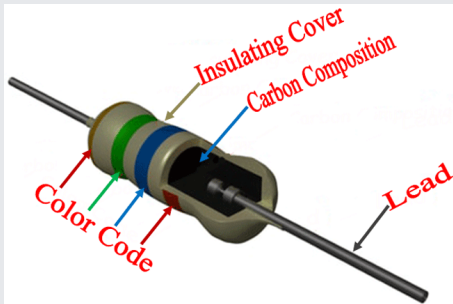


There are mainly **two types of Resistors** used in circuits and laboratories :

① **Wire bound resistors :**

- They are made by winding the wires of an alloy, viz., manganin, constantan, nichrome or similar ones.
- They are **expensive, large**, but their **resistivity is insensitive to temperature**.
- These resistances are typically in the **range** of a fraction of **an ohm to a few hundred ohms**.

Types of Resistors



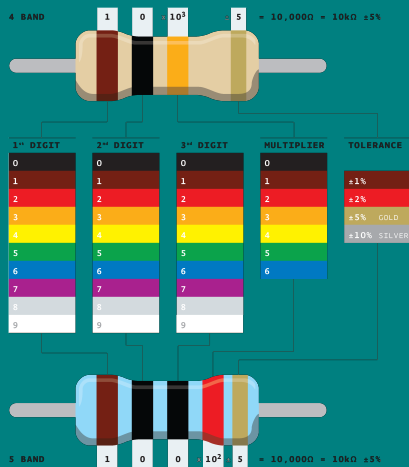
There are mainly **two types of Resistors** used in circuits and laboratories :

① **Wire bound resistors :**

② **Carbon resistors :**

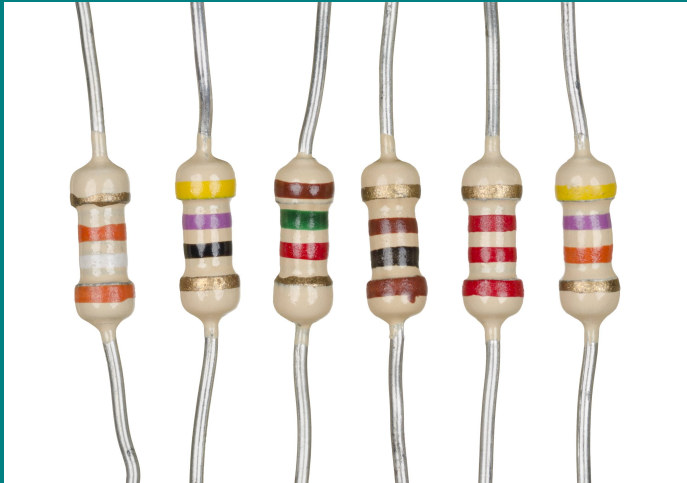
- Made up of **Carbon compounds**.
- Carbon resistors are **compact, inexpensive** and thus used in small electronic circuits.
- Because of their **small size**, their **values** are given using a **colour code**.

Colour Codes for Carbon Resistors



- The first two rings give the first two significant figures of resistance in ohm.
- The third ring indicates the decimal multiplier.
- The last ring indicates the tolerance in per cent about the indicated value. Eg.
 $AB \times 10^C \pm D\% \Omega$
- Short Trick : **B B ROY of Great Britain has Very Good Wife**

Colour Codes for Carbon Resistors



TEMPERATURE DEPENDENCE OF RESISTIVITY

- The **resistivity** of a material is found to be **dependent on the temperature**.
- Different materials do not exhibit the same dependence on temperatures.
- Over a limited range of temperatures(ie., not too large), the **resistivity** of a **metallic conductor** is approximately given by,

$$\rho_T = \rho_0 [1 + \alpha(T - T_0)] \quad (1)$$

where, $\rho_T \implies$ **Resistivity at a temperature T**

$\rho_0 \implies$ **Resistivity at reference temperature T_0 .**

$\alpha \implies$ **Temperature coefficient of resistivity.**

TEMPERATURE DEPENDENCE OF RESISTIVITY

- Multiplying both sides by $\frac{l}{A}$ of eq(1), We get.

$$\rho_T \frac{l}{A} = \rho_0 \frac{l}{A} [1 + \alpha(T - T_0)]$$

$$\boxed{R_T = R_0 [1 + \alpha(T - T_0)]} \quad \because R = \frac{\rho l}{A} \quad (2)$$

where, $R_T \implies$ Resistance at a temperature T
 $R_0 \implies$ Resistance at reference temperature T_0 .
 $\alpha \implies$ Temperature coefficient of resistivity.

TEMPERATURE DEPENDENCE OF RESISTIVITY

- The **dimensions** of α is $[Temperature^{-1}]$ or $[K^{-1}]$
- For **metals**, **Temp. Coefficient of resistivity** (α) is **positive**.
- The above **linear relation** between resistivity (ρ) and Temp. (T) eq(1) does **not hold for all material**.
- Some materials like **Nichrome** (which is an alloy of nickel, iron and chromium), **Manganin and constantan** (used in wire bound standard resistors) exhibit a very **weak dependence of resistivity with temperature**

TEMPERATURE DEPENDENCE OF RESISTIVITY

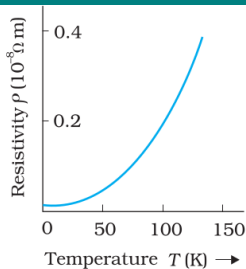


FIGURE 3.9
Resistivity ρ_T of copper as a function of temperature T .

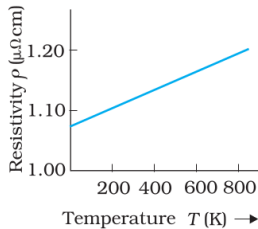


FIGURE 3.10 Resistivity ρ_T of nichrome as a function of absolute temperature T .

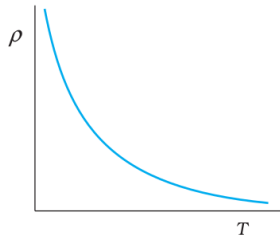


FIGURE 3.11
Temperature dependence of resistivity for a typical semiconductor.

Why resistivity ρ depends on temperature (T)

- **Resistivity** (ρ) of a material is given by-

$$\rho = \frac{m}{ne^2\tau} \quad (3)$$

- ρ thus **depends inversely** both on the **electron no. density** n and on the **relaxation time** τ
- As we **increase temperature**, **average speed of the electrons increases** resulting in **more frequent collisions**. Thus τ **decreases with temperature**.
- **In a metal, resistivity increases with temperature.**
Reason -relaxation time τ **decrease with rise in temperature** but n is **not dependent on temperature** to any appreciable extent.

Why resistivity ρ depends on temperature (T)

- **Resistivity** (ρ) of a material is given by-

$$\rho = \frac{m}{ne^2\tau} \quad (4)$$

- For **insulators and semiconductors resistivity** (ρ) **decreases with temperature.**

Reason - number of conduction electrons per unit volume n increases with temperature. This increase more than compensates any decrease in τ .

Example 3.4

The resistance of the platinum wire of a platinum resistance thermometer at the ice point is $5\ \Omega$ and at steam point is $5.23\ \Omega$. When the thermometer is inserted in a hot bath, the resistance of the platinum wire is $5.795\ \Omega$. Calculate the temperature of the bath.

Solution:

Given :

Resistance (R_0) at ice Point ($T_0 = 0^\circ\text{C}$) = $5\ \Omega$

Resistance (R_{100}) at steam Point ($T_{100} = 100^\circ\text{C}$) = $5.23\ \Omega$

Resistance (R_T) at hot bath (T) = $5.795\ \Omega$

Example 3.4

Solution:

Given :

Resistance (R_0) at ice Point ($T_0 = 0^\circ C$) = 5Ω

Resistance (R_{100}) at steam Point ($T_{100} = 100^\circ C$) = 5.23Ω

Resistance (R_T) at hot bath (T) = 5.795Ω

Step 1 : First we calculate the temp. of coefficient of resistivity α

$$R_T = R_0 [1 + \alpha(T - T_0)]$$

$$R_{100} = R_0 [1 + \alpha(T_{100} - T_0)]$$

$$5.23 \Omega = 5 \Omega [1 + \alpha(100^\circ C - 0^\circ C)]$$

$$5.23 = 5 + 500\alpha^\circ C$$

Example 3.4

Solution:

$$\begin{aligned}\alpha &= \frac{5.23 - 5}{500\text{ }^{\circ}\text{C}} = \frac{0.23}{500\text{ }^{\circ}\text{C}} = 0.00046\text{ }^{\circ}\text{C}^{-1} \\ \alpha &= 4.6 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}\end{aligned}\tag{5}$$

Step 2: Now, using this value of α calculating the temp. of the bath.

$$\begin{aligned}R_T &= R_0 [1 + \alpha(T - T_0)] \\ 5.795\text{ }\Omega &= 5\text{ }\Omega [1 + 4.6 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}(T - 0^{\circ}\text{C})] \\ \frac{5.795}{5} &= 1 + 4.6 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}(T)\end{aligned}$$

Example 3.4

Solution:

$$1.159 = 1 + 4.6 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}(T)$$

$$1.159 - 1 = 4.6 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}(T)$$

$$\frac{0.159}{4.6 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}} = (T)$$

$$T = 0.03456 \times 10^4 \text{ } ^\circ\text{C}$$

$$T = 345.6 \text{ } ^\circ\text{C} \quad (6)$$